Combinatorial Proof

Suppose that there are m elements in one set and n elements in a second set (the two sets are disjoint). Then the total number of ways to choose r elements from the union of the two sets is $\binom{m+n}{r}$.

Another way to choose r elements from the union is to choose $r - k$ elements from the first set and then k elements from the second set, where k is an integer with $0 \leq k \leq r$.

By the product rule, this can be done in $\binom{m}{r-k}\binom{n}{k}$ ways.

Consequently,

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k}.$$